

MARCOS RUIZ MUÑOZ

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PROFESSIONAL PROFILE

Junior Gameplay Programmer with experience in gameplay systems, AI, and UI development in professional indie game production environments.

EDUCATION

Bachelor's degree in Digital Interactive Content

ENTI – Universitat de Barcelona, Spain

2022 – 2026

EXPERIENCE

Intern Gameplay Programmer (Unannounced Project) / July 2025 - May 2026

The Breach Studio

- Implemented AI systems including pathfinding and decision-making logic in Unity.
- Developed gameplay mechanics and systems within a data-driven Scriptable Objects architecture.
- Built adaptive UI components with uGUI, targeting both mobile and pc platforms.
- Integrated Spine skeletal animations into Unity, synchronizing them with gameplay logic.
- Created custom editor tools to streamline the design team's workflow.
- Collaborated remotely with a multidisciplinary team, progressively taking on greater responsibility

KEY PROJECTS

Lyra - Unreal Engine 5 (AI Programmer)

(University Project)

- Designed and implemented a custom locomotion system for two dragon agents using cubic Bézier curves, generating organic and fluid trajectories in real time over NavMesh
- Built a dual-architecture AI system combining a Finite State Machine for physical state management and a Behaviour Tree for cognitive decision-making, decoupled via Blueprint Interfaces
- Developed a spatial analysis system using Navigation Raycasting and Project Point to Navigation to dynamically validate agent trajectories across three different level geometries
- Implemented a centralized avoidance manager coordinating simultaneous navigation of two agents without modifying the base architecture

Unity AI Machine Learning

Ongoing Personal Project (C#)

- Built a reinforcement learning environment in Unity ML-Agents for autonomous tank combat.
- Configured Curriculum Learning and Self-Play training pipelines to optimize learning efficiency and encourage emergent strategies.
- Engineered reward functions, observations, and action spaces to train agents capable of navigation, obstacle avoidance, aiming, and combat.
- Evaluated and iterated on training performance through multiple experiments, improving agent behavior and robustness.

TECHNICAL SKILLS

- **Languages:** C++, C, C#
- **Tools:** Unreal Engine, Unity, Unity ML-Agents, Unity Spine, Jira, Git, Perforce